

SkillForge AI

Platform Architecture, Languages, Tech Stack & Requirements Manual

1. Executive Platform Overview

SkillForge AI is a next-generation engineering career readiness platform and interactive physics-driven skill sandbox. It features a 2D physics gravity simulation engine (Matter.js), 3D WebGL particle/neural fields (Three.js), deterministic diagnostic skill calibrations, real-time role gap analysis, automated resume builders, and an AI interview simulator.

2. Programming & Markup Languages Used & Their Exact Roles

Language / Format	Where & How It Is Used in the Project	Key Responsibilities & Function
JavaScript (ES6+)	- Frontend (React, Zustand, Services) - Backend (Node.js, Express, Middleware)	Core programming language for both client & server runtime. Manages reactive state, physics calculations, algorithms, API endpoints, and authentication.
JSX (React)	All Frontend UI components & pages (<code>`frontend/src/components`</code>, <code>`pages`</code>)	Declarative UI syntax blending HTML structure with JavaScript reactivity, component state, conditional renders, and Framer Motion animation triggers.
HTML5	<code>`frontend/index.html`</code> - Canvas 2D & WebGL 3D canvas mount elements	Defines foundational semantic DOM structure, viewport responsiveness, font preloading, and canvas rendering targets for Matter.js & Three.js.
CSS3 & Tailwind CSS	<code>`frontend/src/index.css`</code> - Tailwind utilities & inline responsive CSS	Neo-brutalist & glassmorphic UI design tokens, box-shadow depth layers, responsive layouts, glowing atmospheric planet pulses, and animation keyframes.
SQL (PostgreSQL) / NoSQL	<code>`backend/config/database.js`</code> <code>`frontend/src/services/firebaseSync.js`</code>	Persistent database storage for user profiles, test logs, domain preferences, and real-time Firestore synchronization of skill matrices.
JSON	<code>`package.json`</code>, REST API payloads, dataset structures	Standard data interchange format for REST API communication, mock datasets (<code>`domainsData.js`</code> , <code>`demoProfile.js`</code>), and package configurations.

3. Complete Framework & Library Breakdown

Layer / Subsystem	Technology / Library	Purpose & Role in SkillForge AI
Frontend Framework	React 18 + Vite	High-performance client application with blazing-fast Hot Module Replacement (HMR) and optimized Rollup production builds.
Physics Simulation	Matter.js (v0.20+)	2D rigid-body engine powering the Earth gravity core, orbital attraction, collision shockwaves, drag/throw mechanics, and Zero-G mode.
3D Graphics & Visuals	Three.js + R3F + Drei	WebGL-rendered 3D hero neural networks, interactive orbital AI interview spheres, and floating background star fields.
Motion & Animations	Framer Motion	Page transitions, gelatinous button bounciness, XP gain toast popups, and smooth modal overlays.
Global State Management	Zustand + Persist	Lightweight reactive state for user auth, diagnostic test scores, active domain selection, and local persistence.
Audio / Sound FX	Web Audio API / Custom Synth	Synthesizes auditory tactile feedback for bubble pops, gravity pulses, test completion, and zero-gravity toggles.
Backend API Server	Node.js + Express.js	RESTful API server with modular routers for auth, skill assessment tracking, AI recommendations, and database access.
Security & Auth	JWT + Bcrypt.js + CORS	Stateless JSON Web Token issuance and verification, salted bcrypt password hashing, and cross-origin resource protection.
Cloud Sync & DB	Firebase Admin / Firestore	Cloud NoSQL database syncing user assessment results and skill scores across sessions in real time.

4. What is Needed to Run & Deploy the Web Platform

To run, maintain, or deploy SkillForge AI from scratch, the following hardware, software, and configuration files are required:

A. System & Runtime Prerequisites

- **Node.js:** Version 18.0.0 or higher (v20+ recommended LTS)
- **Package Manager:** npm (v9+) or Yarn or pnpm
- **Operating System:** Windows 10/11, macOS (12+), or Linux (Ubuntu 20.04+)
- **Web Browser:** Modern browser with Canvas 2D and WebGL acceleration (Chrome, Edge, Firefox, Safari)
- **Git:** Version 2.30+ for repository control and deployment hooks

B. Step-by-Step Execution Guide

Step & Purpose	Terminal Command
1. Install Backend Dependencies	cd backend npm install
2. Start Backend API Server	npm run dev # Starts Express server on http://localhost:5000
3. Install Frontend Dependencies	cd frontend npm install
4. Start Frontend Dev Server	npm run dev # Starts Vite client on http://localhost:5173
5. Build for Production	npm run build # Produces optimized dist/ bundle

C. Required Environment Variables (.env)

Location	Variable Key	Description & Example Value
Backend (.env)	PORT JWT_SECRET FIREBASE_PROJECT_ID DATABASE_URL	Port number (`5000`), secure JWT signature key, optional Firebase credentials, and Postgres connection string.
Frontend (.env)	VITE_API_URL VITE_FIREBASE_API_KEY	Backend base URL (`http://localhost:5000/api`) and Firebase client API key for live database synchronization.

D. Production Hosting & Cloud Architecture

- **Frontend Hosting:** Vercel / Netlify / Cloudflare Pages (Static SPA routing with Vite output).
- **Backend API Hosting:** Render / Railway / AWS EC2 / DigitalOcean App Platform (Node.js container).
- **Database Tier:** Firebase Firestore / Supabase PostgreSQL / MongoDB Atlas.

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